

			-1.51×10^{-5}	
8	(a)	(i)	During braking, when the coil attached to the wheel turns in the magnetic field, the magnetic flux linkage through the coil will be changing.	M1
			According to Faraday's law, since there is a change in magnetic flux linkage in the coil, there will be an induced e.m.f.	M1
			This will produce a current/electricity which can then be stored in the battery and used to power the engine thus improving efficiency.	A1
		(ii)	At lower speeds, the rate of change of magnetic flux linkage will be less,	M1
			hence less induced e.m.f. and less efficiency	A1
		(iii)	Magnetic flux is the product of area and the component of magnetic flux density perpendicular to that area.	B1
		(iv)	$N\Phi = NBA \cos \theta$	M1
		(v)	$\theta = \omega t$	C1
			$E = -\frac{dN\Phi}{dt} = -\frac{dNBA \cos \omega t}{dt} = -\omega NBA \sin \omega t$	M1
			$\max E = \omega NBA = (30 \times 2\pi)(75 \times 0.70 \times 200 \times 10^{-4})$	M1
			$= 200 \text{ V (198)}$	A1
		(vi)	Period doubles and	B1
			height halves	B1
	(b)		Less wear and tear for friction brake pads	B1
			Or smoother braking / prolonged battery life	

	(c)	Heavier vehicles have more mass / inertia, thus their drivers tend to <u>use brakes more often</u> to control the speed of the car hence recovering more energy. OR The regenerative braking system is used in parallel to friction braking system. Heavier vehicles have <u>more momentum/KE/inertia</u> hence require <u>more force</u> to stop. Hence regenerative braking can recover more energy as friction braking might not be able to stop the vehicle completely in that time, as opposed to a lighter vehicle.	B1
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